

RUOMENG DING

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Research Interests: LLM Agents · Adaptive Data Acquisition · LLM Evaluation and Alignment · AI4AI

EDUCATION

- Ph.D. in Computer Science**, [University of North Carolina at Chapel Hill](#) Aug, 2025 - Now
• Advised by [Prof. Zhun Deng](#).
- M.S. in Computer Science**, [Georgia Institute of Technology](#) Aug, 2022 - May, 2025
• GPA: 4.00/4.00, Specialization: [Machine Learning](#).
- B. Eng. in Computer Science and Technology**, [Tianjin University](#) Aug, 2018 - May, 2022
• GPA: 3.75/4.00, Major Rank: 11/169 (6.5%).

PUBLICATIONS

- [COLM 2026] **R. Ding***, Y. Pang*, H. Sun, Y. Wang, Z. S. Wu, and Z. Deng, “Rubrics as an attack surface: Stealthy preference drift in LLM judges,” in *Proceedings of the 3rd Conference on Language Modeling (COLM)*, 2026 (* Equal contribution) [[paper](#)] [[code](#)]
- [ICML 2026] **R. Ding***, T. Gao*, T. P. Zollo, E. Bachmat, R. Zemel, and Z. Deng, “Whom to query for what: Adaptive group elicitation via multi-turn llm interactions,” in *The 43rd International Conference on Machine Learning (ICML 2026)*, 2026 (* Equal contribution) [[paper](#)] [[code](#)]
- [AAAI 2026] **R. Ding**, W. Cheng, M. Shao, and C. Zhao, “Skillgen: Learning domain skills for in-context sequential decision making,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2026 (Oral) [[paper](#)] [[code](#)]
- [ACL 2024] **R. Ding**, C. Zhang, L. Wang, Y. Xu, M. Ma, W. Zhang, S. Qin, S. Rajmohan, Q. Lin, and D. Zhang, “Everything of thoughts: Defying the law of penrose triangle for thought generation,” in *Findings of the Association for Computational Linguistics (ACL)*, 2024 [[paper](#)] [[code](#)]
- [NeurIPS 2024] R. Yang, **R. Ding**, Y. Lin, H. Zhang, and T. Zhang, “Regularizing hidden states enables learning generalizable reward model for LLMs,” in *The Thirty-eighth Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2024 [[paper](#)] [[code](#)]
- [FSE 2023] **R. Ding**, C. Zhang, L. Wang, Y. Xu, M. Ma, X. Wu, M. Zhang, Q. Chen, X. Gao, X. Gao, H. Fan, S. Rajmohan, Q. Lin, and D. Zhang, “Tracediag: Adaptive, interpretable, and efficient root cause analysis on large-scale microservice systems,” in *Proceedings of the 31st ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE)*, 2023 [[paper](#)] [[slides](#)]
- [KDD 2023] L. Wang, C. Zhang, **R. Ding**, Y. Xu, Q. Chen, W. Zou, Q. Chen, M. Zhang, X. Gao, H. Fan, S. Rajmohan, Q. Lin, and D. Zhang, “Root cause analysis for microservice systems via hierarchical reinforcement learning from human feedback,” in *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*, 2023 [[paper](#)]
- [SDM 2025] **R. Ding**, X. Zhao, C. Zhao, M. Shao, Z. Chen, and H. Chen, “Evidence-based out-of-distribution detection on multi-label graphs,” in *Proceedings of the 2025 SIAM International Conference on Data Mining (SDM)*, pp. 588–597, SIAM, 2025 [[paper](#)]
- [VLDB 2023] Y. Chen, C. Zhang, M. Ma, Y. Liu, **R. Ding**, B. Li, S. He, S. Rajmohan, Q. Lin, and D. Zhang, “Imdif-fusion: Imputed diffusion models for multivariate time series anomaly detection,” in *Proceedings of the VLDB Endowment (VLDB)*, 2023 [[paper](#)]

10. [TCYB 2023] T. Li, W. Wang, P. Jiao, Y. Wang, **R. Ding**, H. Wu, L. Pan, and D. Jin, “Exploring temporal community structure via network embedding,” in *IEEE Transactions on Cybernetics (TCYB)*, 2023 [paper]
11. [UDM-AAAI 2023] **R. Ding**, X. Zhao, C. Zhao, and M. Shao, “Detecting multi-label out-of-distribution nodes on graphs,” in *AAAI Workshop on Uncertainty Reasoning and Quantification in Decision Making*, 2023 [paper]

RESEARCH EXPERIENCE

- **Research Intern, Alibaba Tongyi Lab** May, 2026 - Present
 Advised by [Dr. Qianli Shen](#) and [Daoyuan Chen](#) Hangzhou, China (Onsite)
 - Focused on AI4AI. Investigating agentic data selection for supervised fine-tuning (SFT) and reward design for reinforcement fine-tuning (RFT) through a automated agentic framework.
 - Studying the temporal and parallel scaling of agentic data engineering with multi-agent coordination and shared research memory.
- **Research Intern, Microsoft Research** May, 2024 - Aug, 2024
 Advised by [Dr. Minghua Ma](#) and [Dr. Ze Li](#) Redmond, WA (Onsite)
 - Focused on large language model-based multi-agent systems for incident triage.
 - Designed the methodology, conducted experiments, and developed prototypes; presented the work to Microsoft Research and Azure.
- **Research Assistant, University of Illinois Urbana-Champaign** Mar, 2024 - Jul 2024
 Advised by [Dr. Tong Zhang](#), working with [Rui Yang](#) Urbana, IL (Remote)
 - Focus on enhancing the Reward Model’s generalization ability against distribution shifts for Reinforcement Learning from Human Feedback (RLHF). (*NeurIPS 2024, 2nd author*)
 - Responsible for literature review and RLHF experiments. Contribute to paper writing and code repository.
- **Research Intern, Microsoft Research Asia** Nov, 2022 - Aug, 2023
 Advised by [Dr. Lu Wang](#) and [Dr. Chaoyun Zhang](#) Beijing, China (Hybrid)
 - Focus on Boost LLM reasoning with domain knowledge via Monte Carlo Tree Search (MCTS). Responsible for methodology design, experiments, paper writing, and code open-sourcing. (*ACL 2024, 1st author*)
 - Focus on Root Cause Analysis for large-scale microservices systems, leveraging RL and RLHF to detect the root causes of anomalies. The method related to the paper has been deployed in the M365 system. (*KDD 2023, 3rd author; FSE 2023, 1st author*)
- **Research Assistant, Tianjin University** Aug, 2021 - Aug, 2022
 Advised by [Dr. Minglai Shao](#) and [Dr. Wenjun Wang](#) Tianjin, China (Onsite)
 - Focus on Multi-label Out-of-Distribution detection on graphs. Co-advised by [Dr. Xujiang Zhao](#) and [Dr. Chen Zhao](#). (*UDM-AAAI 2022, 1st author*)
 - Focus on dynamic community detection on graphs. Responsible for part of the experiments. (*TCYB 2023*)

HONORS AND FELLOWSHIPS

1. **Doctoral Merit Fellowship**, University of North Carolina at Chapel Hill 2025 - 2026
2. **Merit Scholarship**, Georgia Institute of Technology 2022 - 2023

ACADEMIC SERVICE

- Conference Reviewer: AAAI (2027); CoLM, NeurIPS, ICML, and ICLR (2026); KDD (2025–2026).